

Renal Reference — Polyuria D Dx (With vs Without AKI)

Polyuria D Dx — with AKI vs without AKI

Category	Cause	Core mechanism	Clues
Polyuria WITH AKI	Post-obstructive diuresis	After obstruction is relieved, kidney dumps retained salt/water/urea + tubules temporarily cannot concentrate urine	After Foley/stent/nephrostomy; risk dehydration, hypokalemia, hypotension
Polyuria WITH AKI	Recovery phase of ATN	GFR starts improving before tubules recover reabsorption/concentration	Creatinine stabilizes/improves, then urine output rises
Polyuria WITH AKI	Nephrotoxic nonoliguric ATN	Tubular injury but urine flow is preserved/increased	Aminoglycosides classic; also cisplatin, amphotericin B, ifosfamide, tenofovir
Polyuria WITH AKI	AIN / tubulointerstitial disease	Interstitial inflammation damages tubular concentrating ability	WBCs, WBC casts, sterile pyuria; fever/rash/eosinophilia may be absent
Polyuria WITH or WITHOUT AKI	Osmotic diuresis	Solute pulls water into urine; if severe → volume depletion → prerenal AKI/ATN	Hyperglycemia/DKA/HHS, mannitol, high urea load
Polyuria WITHOUT AKI	Diabetes insipidus	ADH problem → kidney cannot retain free water	Very dilute urine, hypernatremia if no water access
Polyuria WITHOUT AKI	Primary polydipsia	Excess water intake suppresses ADH	Very dilute urine, often low/normal sodium
Polyuria WITHOUT AKI	Diuretics	Forced Na ⁺ /water excretion	Medication history; loop/thiazide use
Polyuria WITHOUT AKI	Hypercalcemia / hypokalemia	Causes nephrogenic DI-like concentrating defect	Electrolyte abnormality + dilute urine
Polyuria WITHOUT AKI	CKD concentrating defect	Damaged kidneys lose ability to concentrate urine early	Nocturia/polyuria can happen before severe GFR decline

1. Post-obstructive diuresis (WITH AKI)

WHAT YOU SEE

Polyuria WITH AKI — Post-obstructive diuresis: relief of obstruction → kidney dumps retained salt/water/urea; tubules temporarily cannot concentrate

WHAT IT ENCODES

After relief of obstruction (Foley for BPH, ureteral stent, percutaneous nephrostomy, stone passage), the kidney offloads retained salt, water, and urea (appropriate component) on top of a transient tubular concentrating defect (pathologic component). Output may exceed 200 mL/h. Risks: dehydration, hypokalemia, hypomagnesemia, hypotension. Replace only ~half to two-thirds of output with isotonic fluid so an appropriate diuresis can self-limit.

BOARD TRAP

WRONG: chasing every mL of urine 1:1 with IV fluid after relieving obstruction. Aggressive matched replacement perpetuates a diuresis that should resolve on its own. Discriminator: separate appropriate (clears retained solute, self-limits) from pathologic (true tubular defect) — replace partially and trend weight, Na, K, Mg, and volume status.

2. Recovery phase of ATN (WITH AKI)

WHAT YOU SEE

Recovery phase of ATN: GFR improves before tubules recover reabsorption/concentration

WHAT IT ENCODES

In recovering ATN the GFR (filtration) returns before the tubules regain reabsorptive/concentrating capacity, and accumulated urea adds an osmotic load → a brisk polyuric phase. Creatinine stabilizes then falls as urine output rises. This phase carries genuine risk of hypovolemia and hypokalemia/hypophosphatemia/hypomagnesemia.

BOARD TRAP

WRONG: easing off monitoring once creatinine starts dropping. The polyuric recovery phase can cause severe volume and K/Mg/PO4 depletion exactly when the team assumes recovery is complete. Discriminator: rising output + falling creatinine after ATN = recovery diuresis — keep replacing fluids/electrolytes and rechecking labs.

3. Nephrotoxic nonoliguric ATN (WITH AKI)

WHAT YOU SEE

Nephrotoxic nonoliguric ATN: tubular injury but flow preserved/increased

WHAT IT ENCODES

Direct tubular toxins necrose cells while urine flow stays preserved/increased. Aminoglycosides are the classic culprit (non-oliguric, Mg/K wasting, onset ~5–10 days, dose once-daily and follow troughs); also cisplatin (Mg wasting), amphotericin B (distal RTA + K/Mg wasting; lipid formulations less toxic), ifosfamide (Fanconi), and tenofovir (proximal/Fanconi). Sediment: muddy-brown granular casts + renal tubular epithelial cells.

BOARD TRAP

WRONG: AKI must be oliguric, so brisk output excludes tubular injury. Nephrotoxic ATN is frequently NON-oliguric — high urine output does not exclude necrosis. Discriminator: granular casts/RTE cells + a nephrotoxin + rising creatinine confirm ATN regardless of output.

4. AIN / tubulointerstitial (WITH AKI)

WHAT YOU SEE

AIN / tubulointerstitial disease: interstitial inflammation damages tubular concentrating ability

WHAT IT ENCODES

Interstitial inflammation impairs the medullary gradient and tubular ADH responsiveness → concentrating defect and polyuria, with subacute AKI. Top drug triggers: PPIs, NSAIDs, beta-lactams, sulfa, allopurinol, rifampin, and immune checkpoint inhibitors. Sediment: WBCs, WBC casts, sterile pyuria; eosinophiluria is unreliable. First step: stop the culprit drug; add steroids if no recovery within ~5–7 days.

BOARD TRAP

WRONG: ruling out AIN because there is no fever, rash, or eosinophilia. The classic triad is INCOMPLETE/absent in the large majority of cases (especially PPI- and checkpoint-inhibitor-induced). Discriminator: subacute AKI + sterile pyuria + WBC casts + a new drug is sufficient to suspect AIN and stop the agent.

5. Osmotic diuresis (WITH or WITHOUT AKI)

WHAT YOU SEE

Osmotic diuresis: solute pulls water into urine; if severe → volume depletion → prerenal AKI

WHAT IT ENCODES

A non-reabsorbed solute load (hyperglycemia in DKA/HHS once filtered glucose exceeds the reabsorptive threshold, mannitol, or a high urea load from catabolism/GI bleed/TPN) holds water in the tubule → solute-rich diuresis. It straddles both columns: mild = polyuria without AKI; severe = volume depletion → prerenal AKI → ATN. Urine osm is not maximally dilute and daily solute excretion is high.

BOARD TRAP

WRONG: treating osmotic-diuresis polyuria as a primary ADH/water-handling defect (chasing DI). The driver is solute, not ADH. Discriminator: high urine osm/solute excretion with an obvious solute source (glucose, mannitol, urea) = osmotic diuresis; correct the solute and replace volume rather than working up DI.

6. Diabetes insipidus (WITHOUT AKI)

WHAT YOU SEE

Diabetes insipidus: ADH problem → kidney cannot retain free water

WHAT IT ENCODES

DI is a free-WATER diuresis from failure of ADH signaling. Central DI = deficient ADH (pituitary surgery, trauma, tumor, infiltrative disease) → responds to desmopressin/DDAVP. Nephrogenic DI = collecting-duct resistance to ADH (lithium, hypercalcemia, hypokalemia, congenital) → does NOT respond to DDAVP; manage with low-salt diet + thiazide ± amiloride (for lithium), and treat the cause. Urine is very dilute (osm often <300, sometimes <100); serum Na rises if water access is restricted.

BOARD TRAP

WRONG: confusing DI with primary polydipsia by sodium. DI gives dilute urine with HIGH/high-normal serum sodium, whereas primary polydipsia gives dilute urine with LOW/low-normal sodium. Discriminator: a water-deprivation test then DDAVP separates central DI (concentrates after DDAVP), nephrogenic DI (no response to DDAVP), and polydipsia (concentrates with deprivation).

7. Primary polydipsia (WITHOUT AKI)

WHAT YOU SEE

Primary polydipsia: excess water intake suppresses ADH

WHAT IT ENCODES

Compulsive/psychogenic water intake (or a low-solute 'tea-and-toast'/beer-potomania pattern) suppresses ADH appropriately → dilute polyuria. Serum sodium is low/low-normal because the excess water dilutes plasma. The kidney is normal and CAN concentrate urine once water intake is restricted.

BOARD TRAP

WRONG: starting DDAVP empirically for dilute polyuria. In primary polydipsia, DDAVP plus continued heavy drinking risks acute hyponatremia. Discriminator: a water-deprivation test concentrates the urine in polydipsia (kidney works) but fails to in DI — confirm before treating.

8. Diuretics (WITHOUT AKI)

WHAT YOU SEE

Diuretics: forced Na⁺/water excretion

WHAT IT ENCODES

Loop (furosemide, bumetanide, torsemide) and thiazide diuretics force a Na⁺/water diuresis. They also invalidate the FENa for AKI workup (forced natriuresis falsely elevates FENa) — use FEUrea <35% instead when diuretics are on board. Surreptitious or unlisted diuretic use is a classic exam twist.

BOARD TRAP

WRONG: launching a full DI/polyuria workup before reviewing the medication list. Diuretics (including OTC/herbal and surreptitious use) reproduce many polyuria patterns. Discriminator: a urine diuretic screen + careful med reconciliation can short-circuit an expensive workup.

9. Hypercalcemia / hypokalemia (WITHOUT AKI)

WHAT YOU SEE

Hypercalcemia / hypokalemia: cause nephrogenic DI-like concentrating defect

WHAT IT ENCODES

Both electrolyte disturbances cause an ACQUIRED nephrogenic DI by impairing collecting-duct ADH responsiveness/aquaporin-2 trafficking and the medullary gradient → dilute polyuria. Hypercalcemia (think malignancy, primary hyperparathyroidism) and chronic hypokalemia are the testable culprits.

BOARD TRAP

WRONG: treating the polyuria itself instead of the electrolyte. These cause a REVERSIBLE acquired nephrogenic DI — correct the calcium or potassium and the concentrating defect/polyuria resolves. Discriminator: dilute polyuria + hypercalcemia or hypokalemia → fix the electrolyte first, don't reach for DDAVP.

10. CKD concentrating defect (WITHOUT AKI)

WHAT YOU SEE

CKD concentrating defect: damaged kidneys lose ability to concentrate urine early

WHAT IT ENCODES

Tubulointerstitial scarring and nephron loss in CKD impair urinary concentration EARLY — often presenting as nocturia/polyuria with a fixed iso-osmotic urine osm (~300) before the serum creatinine rises appreciably. Loss of concentrating ability is one of the earliest functional signs of chronic kidney damage.

BOARD TRAP

WRONG: dismissing new nocturia/polyuria as benign aging or BPH because the creatinine is still normal. Loss of concentrating ability is an EARLY CKD sign that PRECEDES a measurable creatinine rise. Discriminator: persistent nocturia with iso-osmotic urine warrants checking for underlying CKD/tubulointerstitial disease.